

POLARIS DC-1 — BASIS OF DESIGN

Document Ref: PDC-BOD-001 Rev A

Project: Polaris DC-1 — Aviapolis Data Campus, Phase 1, Vantaa, Finland

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Date: 20 February 2026

Status: Design Document — basis for PDC-DWG-* drawings, PDC-SPEC-* specifications and PDC-BOQ-001

Synthetic demonstration data for the White Hat IQ platform. All parties, values and figures are fictional.

1. Purpose and design intent

This Basis of Design (BOD) sets out how NordBuild Oy proposes to satisfy the Employer's Requirements (PDC-ER-001 Rev C) for the Polaris DC-1 facility. It is the single design-intent narrative from which the drawing set, technical specifications and Bill of Quantities are derived; every area, grid dimension, equipment rating and count stated below is intended to be carried forward unchanged into those downstream

documents. Where this BOD states a figure more precisely than the Employer's Requirements (e.g. a specific transformer rating within the required N+1 topology), that figure is NordBuild's proposed means of compliance and remains subject to the Engineer's design review under the S1–S3 submission stages.

1.1 Design intent

The facility is designed as four independently commissionable 2.5 MW data halls sharing common primary infrastructure (110/20 kV substation, standby generation, cooling spine), delivered in four Sections so that each hall can be handed over and brought into live operation without further construction work to the infrastructure it depends on, and without disturbing the operation of any Section already taken over. This single requirement — sectional completion against shared infrastructure — drives the site layout, the structural grid, the electrical topology and the commissioning sequence set out below.

1.2 Design standards

Discipline	Standard
Statutory / planning	Land Use and Building Act (132/1999); National Building Code of Finland; Rescue Act
Structural	Eurocodes EN 1990–1999 with the Finnish National Annexes; snow and wind per the Finnish national annex, Vantaa location
Electrical	SFS 6000 (low voltage); SFS 6001 (high voltage)
Data centre resilience	Uptime Institute Tier III criteria, adopted as the design benchmark (certification not pursued)
Data centre facility standards	EN 50600 series (facility, power, environmental control, telecom cabling), used as the design and PUE-calculation reference (EN 50600-4-2)
Fire	National Building Code fire chapters; EI 120 compartmentation between halls per the Employer's Requirements

Where these standards conflict (e.g. a Tier III concurrent-maintainability expectation exceeding an Eurocode-minimum member size), the more onerous requirement governs.

2. Site layout strategy

2.1 Grid convention

A single primary structural and setting-out grid is used across the whole 4.8 ha plot so that every building, drawing and later BOQ line references the same coordinates:

- **Lettered gridlines A–N**, spaced at **12 m** centres, running north–south (the primary frame bay spacing) — 13 bays, 144 m overall.
- **Numbered gridlines 1–24**, spaced at **18 m** centres, running east–west (the primary frame span direction) — 23 bays, 414 m overall.

All plan dimensions in this BOD and in the PDC-DWG-* set are given as gridline ranges against this convention (e.g. "Hall 1: gridlines B–E / 2–6").

2.2 Arrangement

The plot is long east–west and is developed as a single build-line of four halls plus supporting infrastructure, arranged so that Sections 2–4 can be constructed without re-entering the footprint of an operating Section:

Zone	Gridlines (approx.)	Rationale
Gatehouse and site access	A–C / 1–2	At the single Tikkurilantie access point; controls all site traffic before it reaches any operating hall
Primary substation & switchgear building	A–D / 3–7	Adjacent to the 110 kV termination on the site's north side, upstream of every hall, built in Section 1
Generator compound (enclosed)	A–D / 19–24	East end, away from the north-east residential boundary noise-sensitive receptor; built in Section 1 sized for the full site
Hall 1	E–H / 2–6	West end — shallowest excavation/blasting risk, built first, becomes the "live" reference for later halls
Hall 2	E–H / 8–12	
Hall 3	E–H / 14–18	
Hall 4	E–H / 19–23	East end — coincides with the area requiring blasting, deliberately sequenced last so rock excavation risk sits where it has the most float
Shared MEP spine & mezzanine	I–K / 2–23	Single service corridor running the full building length behind all four halls, so each hall's plant connects to common busway/pipework risers without crossing another hall
Chiller / dry-cooler yard	L–N / 2–23	External yard along the full spine, screened from the north-east residential boundary, split into four hall-aligned bays plus a shared backup-chiller zone
Admin / NOC block	I–K / 20–23, two-storey	East end adjacent to Hall 4 and the gatehouse route, built in Section 4

Fire/service breaks of one bay (18 m) are left between each hall on the numbered grid, both to satisfy the EI 120 compartmentation strategy at the envelope and to give each Section a physical hoarding line during construction.

2.3 Construction logistics consequence

Because Sections are built strictly west-to-east along the shared spine, the site logistics plan uses a single one-way haul road along gridline N (outboard of the cooling yard), with the active laydown and site-office compound migrating eastward one hall at a time as each Section is handed over. Temporary hoarding is erected on the numbered gridline immediately east of the most recently completed hall (e.g. gridline 7 after Hall 1 handover) before work starts on the next Section, and every method statement for work adjacent to an operating hall (per the Employer's Requirements, Section 7) references this same hoarding line. The shared MEP spine and cooling yard are built progressively in Section-sized increments along their full length so that each hall's services are complete at its own handover, without exposing the spine serving an operating hall to open construction.

3. Architectural

3.1 Area schedule

This area schedule follows the Employer's Requirements area schedule (PDC-ER-001 §2.1) line for line and reconciles to the same 14,450 m² total GFA:

Area	GFA (m ²)	Notes
Data Hall 1	1,650	1,250 m ² white space + 400 m ² electrical/mechanical support
Data Hall 2	1,650	as Hall 1
Data Hall 3	1,650	as Hall 1
Data Hall 4	1,650	as Hall 1
Shared MEP spine and mezzanine	2,900	Pipework, busway, UPS and battery rooms serving all four halls
Primary substation and switchgear building	1,100	110/20 kV, dual transformers
Generator compound (enclosed)	950	8 × 2.5 MVA
Administration and NOC block	1,250	Two-storey, offices, NOC, welfare
Security gatehouse and loading	350	
Circulation, plant risers and structural void allowance	1,300	Distributed across halls, spine and substation; closes the gross/net gap in the ER summary table
Total GFA	14,450	Matches PDC-ER-001 §2.1 exactly

Each data hall's gross structural footprint on the primary grid is 4 bays × 2 bays (72 m × 24 m = 1,728 m²); the 1,650 m² net functional area (1,250 m² white space + 400 m² support) is the footprint less external wall thickness and structural zones at the perimeter.

3.2 Clear heights

- White space: **4.2 m minimum clear height** under the lowest service (busway, cable tray or pipe), achieved by setting the underside of primary roof steel at 6.6 m above finished floor level and running all overhead containment within a coordinated 2.4 m service zone.
- Electrical/mechanical support rooms: 4.5 m clear minimum, driven by switchgear and UPS cabinet heights.
- Admin/NOC block: 2.7 m clear per floor, standard commercial fit-out.

3.3 Envelope build-up (indicative, to achieve the ER U-values)

Element	Build-up	U-value
Wall	150 mm precast concrete inner leaf + 200 mm PIR insulation ($\lambda \approx 0.022$ W/mK) + 50 mm ventilated cavity + profiled metal rainscreen outer leaf	0.17 W/m ² K
Roof	Structural steel deck + vapour control layer + 300 mm PIR insulation + single-ply membrane, with a reinforced zone at 150 mm centres coordinated for PV mounting rail fixings	0.09 W/m ² K
Personnel/loading doors	Insulated sectional or personnel doors	1.0 W/m ² K
Airtightness	Taped membrane laps and gasketed panel joints throughout, tested to $q_{50} \leq 1.0$ m ³ /(h·m ²)	—

3.4 Fire compartmentation

Each data hall is a discrete fire compartment separated from its neighbours by precast concrete panels achieving not less than **EI 120**, continuous from foundation to the underside of the roof deck, with all penetrations (busway, pipework, cable routes between the hall and the shared MEP spine) fire-stopped to the same rating. The shared MEP spine is sub-compartmented at each hall's connection point so that a single hall's plant room cannot compromise the spine serving the other three halls. White-space suppression (inert gas or water mist) is treated as a **fire-engineering submission item**, not a deemed-to-satisfy solution — the choice between the two is made at S2 detailed design and is subject to authority approval before it is fixed in the specification (PDC-SPEC-*, fire discipline).

3.5 Finishes strategy

Area type	Floor	Walls	Ceiling
White space	Sealed, static-dissipative coating on the structural slab (no raised floor)	Painted precast, sealed joints	Exposed structure, painted, overhead containment coordinated above
Electrical/mechanical support and battery rooms	Sealed concrete, chemical-resistant in battery rooms	Painted precast/blockwork	Exposed structure
Admin / NOC	Carpet tile (office), raised access floor in the NOC for cabling	Plasterboard, painted	Suspended acoustic tile
Gatehouse	Sealed concrete / vinyl	Painted blockwork	Suspended tile

4. Structural

4.1 Frame strategy

The primary structure is a **steel braced-portal frame on the 12 m × 18 m grid** defined in §2.1: columns at 12 m centres along the lettered gridlines, primary rafters/trusses spanning 18 m between numbered gridlines. Halls are framed as clear-span single-storey structures (no internal columns within the white space) to keep containment and future rack layout unconstrained; the shared MEP spine and the admin/NOC block use the same grid but with an intermediate column line to suit the mezzanine and second floor respectively. Primary columns are UC sections, rafters/trusses are fabricated lattice trusses over the 18 m span to keep depth within the 6.6 m underside-of-steel target, and horizontal bracing bays are located at the gridline intersections adjacent to each fire-compartment wall so that no bracing bay crosses a compartment line.

4.2 Foundations

Ground investigation (PDC-SS-001) records bedrock rising from approximately 9 m below existing ground level in the west of the site to approximately 4 m in the east, with glacial till above rock throughout. The foundation strategy responds directly to this gradient, and to the sectioning set out in §2.2:

- **West zone (Hall 1, primary substation, gatehouse — gridlines 1–7):** reinforced concrete pad footings bearing on compacted till, no blasting required.
- **Central zone (Hall 2, Hall 3 — gridlines 8–18):** pad footings on till where cover to rock exceeds approximately 6 m, transitioning to piers founded directly on rock where cover is shallower, confirmed hole-by-hole during construction against the geotechnical report.
- **East zone (Hall 4, generator compound — gridlines 19–24):** direct-to-rock or rock-socketed piers; controlled blasting required per PDC-SS-001, subject to the vibration limits and neighbour notification set out in PDC-PP-001.

Because the east zone (heaviest blasting exposure) coincides with Hall 4 and the generator compound — the last Section, due Month 22 — blasting productivity risk sits where the programme has the most float; the generator compound's Section 1 delivery date is the exception and is flagged as a programme risk in §10.

4.3 Slab and roof loading

- Ground floor slab: ground-bearing reinforced concrete, designed for a **25 kN/m² uniform imposed load**, FM2 flatness class throughout the white space and support areas (no raised floor).
- Roof: insulated warm roof (§3.3) designed for the Finnish national annex snow and wind loading for Vantaa, plus an additional **0.75 kN/m² uniform allowance for PV panels and mounting rail**, plus point load allowances at rooftop plant curbs (dry-cooler and AHU positions) coordinated with the mechanical layout in §6.
- Design life: 50 years for the primary structure, 25 years for the envelope, per the Employer's Requirements.

4.4 Quantities (BOD-stage estimate, for BOQ development)

Item	Approximate quantity	Basis
Structural steel (frame, bracing, secondary steel)	≈ 830 tonnes	≈ 65 kg/m ² over the single-storey steel-framed areas (halls, spine, substation, generator compound, gatehouse — 11,900 m ²) plus ≈ 40 kg/m ² over the two-storey admin/NOC block (1,250 m ² footprint)
Precast concrete wall panels	≈ 404 panels	Panel 3.0 m wide × 7.0 m tall (21.0 m ²) with joints at 3.0 m centres — 256 panels to the four data halls (perimeter 192 m each) and 148 panels to the spine, substation and generator enclosure, totalling ≈ 8,480 m ² of precast wall area

Both figures are BOD-stage design estimates for early BOQ development and are expected to be refined at S2 detailed design.

5. Electrical

5.1 Topology

Incoming 110 kV terminates within the primary substation building (Contractor's battery limit — the network operator's line and its own terminations are excluded). Two **110/20 kV transformers**, each rated **16 MVA**, step down to a 20 kV site ring; the transformers are sized in N+1 configuration so that either transformer alone can carry the full site peak load (§5.4). The 20 kV ring runs along the shared MEP spine (gridlines I–K) and feeds a dedicated hall substation at each of the four halls, stepping down to 400 V for distribution to the white space via overhead busway, sized to the per-hall load build-up in §5.4. Earthing is a common bonding network, TN-S throughout, with a single site earth reference bar in the primary substation building.

5.2 UPS and batteries

Each hall has its own static, double-conversion UPS system configured **N+1** with **4 × 1.0 MVA modules** (3 duty + 1 standby, giving 3.0 MVA firm capacity against a required ≈ 2.63 MVA at the hall's 2.5 MW IT load and an assumed 0.95 output power factor), sized for **5 minutes autonomy at full load**. Batteries are **lithium-ion (LFP chemistry)**, selected for footprint within the 400 m² per-hall support area and for cycle life under the frequent N+1 module-test regime; the associated fire strategy (per the Employer's Requirements) comprises a dedicated fire-rated battery room per hall, off-gas/thermal-runaway detection ahead of the aspirating smoke system, and physical segregation between UPS modules so that a single-module event cannot propagate.

5.3 Standby generation

Eight **2.5 MVA** diesel generators (20 MVA installed) are located in the enclosed compound at the east end of the site (§2.2), sized N+1 for the full site load including cooling: seven units running (17.5 MVA firm) cover the site's peak requirement of ≈ 16.06 MVA (§5.4), with the eighth as standby. Automatic transfer to generation completes within 10 seconds of loss of incoming supply, with the UPS carrying the load through the transfer; generator load acceptance is demonstrated at full block load during L4/L5 commissioning (§9). Fuel storage is sized for **48 hours at full site load**, banded, with a fuel polishing system serving the compound. The generator compound and its fuel storage are built in Section 1, sized for the full 10 MW site from the outset, per the Employer's Requirements.

5.4 Per-hall load build-up (design/sizing basis)

Electrical infrastructure (transformers, generators, UPS) is sized to the **PUE guarantee ceiling of 1.25** (a 25% overhead on IT load) as a conservative design-sizing basis, distinct from the lower annualised PUE this design is expected to achieve in operation (§7, driven by the high proportion of free-cooling hours).

Item	Per hall (MW)	× 4 halls (MW)
IT load	2.500	10.000
Cooling (design-sizing allowance, 18% of IT)	0.450	1.800
Electrical losses — transformers, UPS, distribution (5% of IT)	0.125	0.500
Lighting and small power, white space + hall support (2% of IT)	0.050	0.200
Total per hall / halls subtotal	3.125	12.500

Site-wide reconciliation against the 16 MVA-class primary infrastructure:

Item	MW
Four halls (from above)	12.50
Admin/NOC block, external lighting, security and site loads (outside the PUE boundary)	0.35
Total site peak (active power)	12.85
Apparent power at assumed 0.95 site power factor	13.53 MVA
Firm capacity, dual 16 MVA transformers (N+1, either alone)	16.00 MVA
Headroom	≈ 2.5 MVA (≈ 18%) — growth, diversity and non-coincident peak allowance

The same 12.85 MW total site peak, divided by an assumed 0.8 generator power factor, gives a required 16.06 MVA of standby generation — the basis for the seven-duty/one-standby generator configuration in §5.3.

6. Mechanical

6.1 Cooling strategy

Cooling is **indirect adiabatic free cooling**, led as the primary mode, with chilled water as backup. The strategy is selected specifically because Vantaa's climate — design conditions of +29 °C dry bulb / +19 °C wet bulb in summer and −29 °C in winter, per the Employer's Requirements — allows the indirect adiabatic plant to meet the 24 °C ± 1.5 K rack-inlet requirement without mechanical (chiller) assistance for the great majority of the year. Outside air is used directly through the indirect heat exchange coil whenever ambient wet-bulb conditions allow the required supply temperature to be met; adiabatic spray assist extends this free-cooling window into warmer conditions by depressing the effective entering air temperature; chilled water backup (served from a shared backup-chiller zone in the cooling yard, §2.2) covers the small number of

hours each year when neither mode alone can meet the setpoint. On this basis the design targets **≥ 8,300 free-cooling hours per year** (≈ 95% of the year), with adiabatic and mechanical assistance covering the remaining ≈ 460 hours during the warmest summer conditions.

6.2 Water use

Adiabatic make-up water is metered and capped at **12,000 m³/year site-wide**, which is the controlling water-use constraint: at the design annual IT energy of 10 MW × 8,760 h = 87,600 MWh, the 12,000 m³ cap gives an effective **WUE of ≈ 0.137 L/kWh**, well inside the 0.25 L/kWh guarantee. The adiabatic system is therefore sized and controlled to the 12,000 m³ cap as the binding limit, not to the 0.25 L/kWh guarantee value, which it comfortably beats as a consequence.

6.3 CRAH, containment and pipework

Each hall has its own bank of CRAH units (indirect-adiabatic air handling units) configured **N+1**, sized so that any single unit can be removed for maintenance without loss of the hall's cooling capacity or IT load (concurrent maintainability). Air management is cold-aisle containment with overhead supply from the

CRAH units, coordinated within the 2.4 m overhead service zone described in §3.2. Chilled-water backup pipework is pre-insulated, welded and pressure-tested, run within the shared MEP spine to a dedicated riser at each hall, with glycol concentration set to the Finnish winter design condition (−29 °C) to prevent freeze risk in any external or unheated run.

6.4 Waste-heat recovery

A waste-heat recovery plant, tapping the hall-side of the chilled-water/adiabatic system, delivers **not less than 4 MW thermal at not less than 70 °C** flow temperature to the site boundary connection chamber,

available whenever site IT load exceeds 40%, per the Employer's Requirements and PDC-DH-001. The recovery

plant is arranged as a parallel offtake from the cooling yard's backup-chiller zone rather than in series with any hall's primary cooling path, so that a fault or isolation of the heat-recovery plant cannot remove cooling capacity from an operating hall — preserving both the N+1 cooling redundancy and the PUE guarantee,

as required. District-heating offtake is only usable, and only tested at full capacity, during the Finnish heating season (broadly October–April); this is carried forward as a commissioning-programme risk in §10.

7. PUE build-up

The annualised PUE calculation below follows EN 50600-4-2 and is weighted by the free-cooling / mechanical-

assist hour split from §6.1, at the full 10 MW design IT load:

Component	Basis	Average power (MW)
IT load	10.000 MW, continuous	10.000
Cooling — free-cooling mode	8,300 h/year at 12% of IT load (fans, adiabatic pumps only) → $1.200 \text{ MW} \times (8,300/8,760)$	1.137
Cooling — mechanical-assist / chilled-water backup mode	460 h/year at 55% of IT load (chillers running) → $5.500 \text{ MW} \times (460/8,760)$	0.289
Cooling subtotal (annualised average)		1.426
Electrical losses (transformers, UPS, distribution)	5% of IT load	0.500
Lighting and small power (white space + hall support)	2% of IT load	0.200
Total facility power		12.126
Annualised PUE	$12.126 / 10.000$	1.213

This is below the ≤ 1.25 guarantee with a margin of ≈ 0.037 , consistent with electrical infrastructure being sized to the 1.25 ceiling (§5.4) while operating in a climate that delivers substantially more free-cooling hours than the 8,300 h/year design minimum in most years.

8. Fire, security and controls

8.1 Fire

Detection is aspirating smoke detection throughout the white space and an addressable system elsewhere. Suppression is sprinkler to the National Building Code in all areas other than the white space; white-space suppression (inert gas or water mist) is a fire-engineering submission item requiring authority approval before it is fixed at S2 (§3.4) — it is not treated as settled by this BOD. Compartmentation is EI 120 between halls (§3.4), with dry risers provided where required by the authority.

8.2 Security

Security is arranged in five layers, from the site boundary inward:

1. Perimeter fence and intrusion detection around the full 4.8 ha plot.
2. ANPR vehicle gate at the single Tikkurilantie access point (§2.2).
3. Mantrap at building reception, admin/NOC block.
4. Section-level access-controlled circulation along the shared MEP spine, restricting movement between an operating Section and an under-construction Section.
5. Hall-level and cage-level card-plus-biometric access to the white space itself, integrated with the Tenant's credential database at each Section's handover.

CCTV covers the perimeter, gatehouse, spine circulation and each hall entry, with 90-day retention.

8.3 Controls

The BMS covers all mechanical and electrical plant on an **open protocol (BACnet/IP)**, avoiding a single-vendor lock-in across four halls built and commissioned up to eight months apart. The DCIM sits above the BMS, providing per-hall and per-feed power, cooling and environmental reporting (per the metering strategy in §5.1) with 24 months of historical trending retained. The alarm hierarchy is agreed with the Employer before commissioning begins on each Section, and every setpoint, sequence and cause-and-effect matrix is documented and witnessed as part of L3/L4 commissioning (§9) before that Section's alarms are handed to the Employer with no unassigned critical alarms outstanding.

9. Commissioning strategy

Commissioning follows the five-level structure (L1–L5) per the Employer's Requirements, applied Section by Section:

Level	Activity	Section-level application
L1	Factory acceptance testing (witnessed for HV switchgear, UPS, generators, chillers)	Ahead of delivery, timed to each Section's equipment need date
L2	Site delivery and static inspection	On arrival at the relevant hall or the shared substation/generator compound
L3	Individual component start-up and functional test	Per system, per hall, as each hall's plant is completed
L4	Integrated systems testing, including black-building and failure-mode tests	Per Section, once all of that Section's systems have passed L3
L5	Integrated load-bank testing to full Section load for 24 hours continuous	Per Section — see below for the Section 1 exception

Section 1 carries the shared infrastructure sized for the full 10 MW (primary substation, both transformers, all eight generators, the full-length cooling spine and its backup-chiller zone). Hall 1's own IT load is only 2.5 MW, so Section 1's L5 test cannot rely on Hall 1's real demand to prove the shared plant. Section 1 L5 testing therefore uses temporary load banks connected at the hall substations to load the primary substation, generator compound and shared cooling spine to their full **10 MW rated capacity** (not just Hall 1's 2.5 MW) for the 24-hour continuous test, demonstrating that Sections 2–4 can be handed over later without any further work to shared infrastructure. Sections 2–4 L5 tests then load-bank only the incremental hall being handed over, since the shared infrastructure was already proved at full capacity in Section 1.

Tests after Completion (the 12-month metered PUE demonstration, §7) run against the whole operating facility as each Section is added, with the annualised PUE calculation re-based on actual occupied IT load once all four halls are live.

10. Key design assumptions and risks

#	Assumption / risk	Consequence if wrong
1	Grid operator will confirm a firm 110 kV energisation date early enough to support Section 1's L4/L5 commissioning window (target ≈ Month 13)	If the energisation date slips, Section 1 L4/L5 (and the temporary-load-bank test in §9) cannot start on programme even if all site works are complete
2	District-heating offtake can only be connected and tested at full capacity during the Finnish heating season (≈ October–April)	Section 1 is due ≈ Month 14 (April 2027), at the tail of the heating season; any slippage into summer defers full-capacity heat-recovery testing to the following heating season, independent of construction progress
3	Blasting productivity in the eastern third (Hall 4, generator compound foundations) is assumed at a standard controlled-blasting rate from PDC-SS-001	Lower-than-assumed rock-breaking productivity directly affects Hall 4's foundation start, though it sits in Section 4 (Month 22) where float is greatest; the generator compound is also in the east and is required in Section 1, where there is materially less float
4	Long-lead HV switchgear, UPS modules and generators are ordered against the design ratings in §5 at contract award, with no design change after order	A rating change after order (e.g. transformer size) re-triggers the long-lead procurement cycle, which is on the critical path for Section 1
5	Tenant fit-out access to Hall 1 begins Month 12, two months before Section 1 completion (Month 14)	Overlap between the Contractor's own Section 1 completion works and Tenant fit-out activity requires the live-environment method statement (per the Employer's Requirements) to also function as a live-construction method statement in the reverse direction — protecting the Contractor's own completion works from a Tenant already on site
6	Ground conditions in the central zone (Hall 2/Hall 3, gridlines 8–18) follow the till-over-rock gradient recorded in PDC-SS-001 without significant local variation	Local rock highs or lows in the central zone would require foundation type to be confirmed pier-by-pier during construction rather than by zone, as currently assumed

End of PDC-BOD-001 Rev A.

SYNTHETIC DEMONSTRATION DATA — NOT A REAL PROJECT